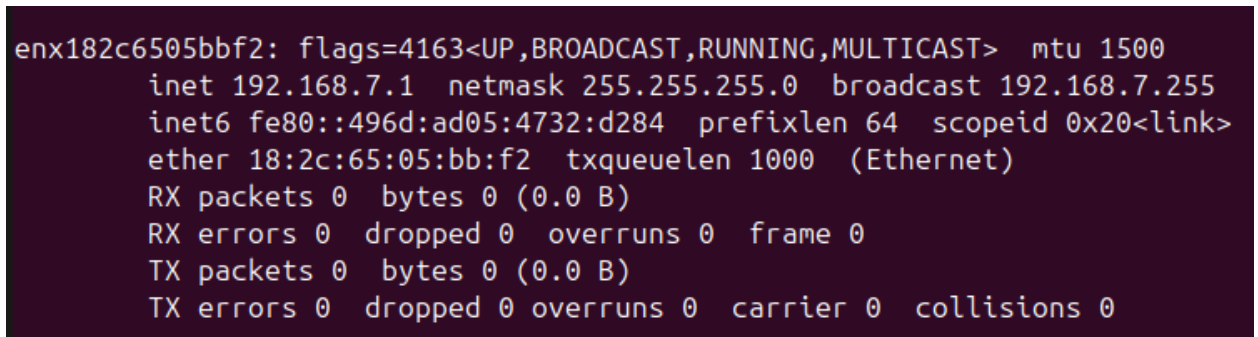
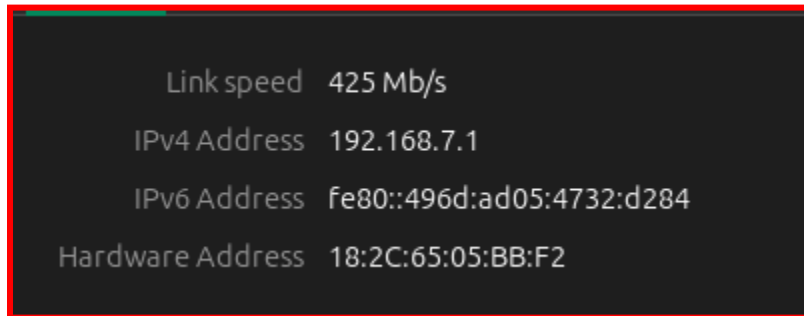


Beaglebone- black Internet Over USB (For Linux Host)

1. Make a USB ethernet profile first. Upon clicking “add ethernet connection” your host will create a profile for your PC’s USB interface and will assign it an IP address. Example: 192.168.7.1 (in my case).



2. From the above step:
> IPv4 checked as Automatic DHCP.
> IPv6 checked as Automatic.

Note: Keep them as it is unless you have some special needs assigning ip addresses manually.

3. Now ping your host. **Note that your Default Gateway of the BeagleBone Black depends on whatever is set by the DHCP or valid Manual IPs that you assign to your USB interface. Refer to the picture above in the 1st step.**

Example:-

```

debian@BeagleBone:~$ ping 192.168.7.1
PING 192.168.7.1 (192.168.7.1) 56(84) bytes of data.
64 bytes from 192.168.7.1: icmp_seq=1 ttl=64 time=4.54 ms
64 bytes from 192.168.7.1: icmp_seq=2 ttl=64 time=2.28 ms
64 bytes from 192.168.7.1: icmp_seq=3 ttl=64 time=2.34 ms
64 bytes from 192.168.7.1: icmp_seq=4 ttl=64 time=2.27 ms
64 bytes from 192.168.7.1: icmp_seq=5 ttl=64 time=2.28 ms
64 bytes from 192.168.7.1: icmp_seq=6 ttl=64 time=2.30 ms
64 bytes from 192.168.7.1: icmp_seq=7 ttl=64 time=2.30 ms
64 bytes from 192.168.7.1: icmp_seq=8 ttl=64 time=2.29 ms
64 bytes from 192.168.7.1: icmp_seq=9 ttl=64 time=2.27 ms
64 bytes from 192.168.7.1: icmp_seq=10 ttl=64 time=2.29 ms
64 bytes from 192.168.7.1: icmp_seq=11 ttl=64 time=2.27 ms
64 bytes from 192.168.7.1: icmp_seq=12 ttl=64 time=2.31 ms
64 bytes from 192.168.7.1: icmp_seq=13 ttl=64 time=2.32 ms

```

- Now, switch to your target or BeagleBone Black.

Navigate to `/etc/systemd/resolved.conf` and uncomment the following:

#DNS=

#FallbackDNS=

Then add the following:

DNS=8.8.8.8 1.1.1.1

FallbackDNS= 8.8.4.4 9.9.9.9

And, restart the systemd-resolved service:

sudo systemctl restart systemd-resolved

Note: In recent versions of the linux, if you don't make the changes in the system-resolved service will generate an empty nameserver list every time you boot your Host.

- Then, set the IP address of the USB interface as the default gateway of your BeagleBone Black. Run the following command:

sudo ip route replace default via <ip-of-the-USB-interface> dev <USB-interface>

Example:- sudo ip route replace default via 192.168.7.1 dev usb0

Let's verify

```

debian@BeagleBone:~$ ip route
default via 192.168.7.1 dev usb0
192.168.7.0/24 dev usb0 proto kernel scope link src 192.168.7.2
debian@BeagleBone:~$

```

Note: You need to run the above command every time your target or BeagleBone Black resets.

- Navigate to the `/etc/sysctl.conf` and uncomment **net.ipv4.ip_forward** and set it to 1
Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1

And also the add the following

```
net.ipv4.conf.all.forwarding = 1
```

```
net.ipv4.conf.default.forwarding = 1
```

```
net.ipv4.conf.<usb-interface-name>.forwarding = 1
```

```
net.ipv4.conf.<ethernet-interface or wlan-interface name>.forwarding = 1
```

Note: Check the ifconfig for the interface names in your PC

7. Now the run the following command in your host:

```
> echo 1 > /proc/sys/net/ipv4/ip_forward
```

```
> iptables -t nat -A POSTROUTING -o <ethernet-interface or wlan-interface name> -j MASQUERADE
```

```
> sudo iptables -A FORWARD -i <usb-interface-name> -o <ethernet-interface or wlan-interface name> -j ACCEPT
```

```
> sudo iptables -A FORWARD -i <ethernet-interface or wlan-interface name> -o <usb-interface-name> -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

Check the ifconfig for the interface names in your PC

Note: You need to run these above commands every time your host resets.

8. Now let's ping google from the target.

```
debian@BeagleBone:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=38.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=37.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=38.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=38.8 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 37.736/38.352/38.774/0.391 ms
debian@BeagleBone:~$ ping www.google.com
PING www.google.com (142.251.151.119) 56(84) bytes of data.
64 bytes from 142.251.151.119: icmp_seq=1 ttl=116 time=37.0 ms
64 bytes from 142.251.151.119: icmp_seq=2 ttl=116 time=37.7 ms
64 bytes from 142.251.151.119: icmp_seq=3 ttl=116 time=39.3 ms
64 bytes from 142.251.151.119: icmp_seq=4 ttl=116 time=38.6 ms
64 bytes from 142.251.151.119: icmp_seq=5 ttl=116 time=38.6 ms

--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 36.968/38.227/39.251/0.797 ms
debian@BeagleBone:~$
```

Now, it should work as expected or else better one can buy an ethernet cable without going through all these.